vibrational motion. As the uniform vibrations we are currently focusing on have energies that are inversely dependent on the radius, they are proportional to whole-number multiples of the reciprocal of the radius—1/R—which in this case is one-tenth of the Planck length. We call this whole number multiple the vibration number.

2

As you can see, this situation is very similar to what we encountered on Wall Street, with the winding and vibration numbers being direct analogs of the shares held in the two companies, while R and 1/R are the analogs of the closing prices per share in each. Now, just as you can easily calculate the total value of your investment from the number of shares held in each company and the closing prices, we can calculate the total energy carried by a string in terms of its vibration number, its winding number, and the radius. In Table 10.1 we give a partial list of these total energies for various string configurations, which we specify by their winding and vibration numbers, in a Garden-hose universe with radius R = 10.

A complete table would be infinitely long, since the winding and vibration numbers can take on arbitrary whole-number values, but this representative piece of the table is adequate for our discussion. We see from the table and our remarks that we are in a high-winding-energy/low-vibration-energy situation: Winding energies come in multiples of 10, while vibrational energies come in multiples of the smaller number 1/10.

Now imagine that the radius of the circular dimension shrinks, say, from 10 to 9.2 to 7.1 and on down to 3.4, 2.2, 1.1, .7, all the way to .1 (1/10), where, for our present discussion, it stops. In this geometrically distinct form of the Garden-hose universe we can compile an analogous table of string energies: Winding energies are now multiples of 1/10 while vibration energies are multiples of its reciprocal, 10. The results are shown in Table 10.2.

At first glance, the two tables might appear to be different. But closer inspection reveals that although arranged in a different order, the “total energy” columns of both tables have identical entries. To find the corresponding entry in Table 10.2 for a chosen entry in Table 10.1, one must simply interchange the vibration and winding numbers. That is, vibration and winding contributions play complementary roles